Riyanna Wilson

IT 214-DL1

Happy Learning Database ERD and Description

The following is an in-depth descriptions and image of the ERD of the Happy Learning Database Kindergarten school. The following design includes descriptions of all information stored including students, parents, classes, and teachers.



Conceptual Analysis

The main entities are class, teacher, parent, and student. The teachers teach the classes; the students are assigned to a class. The student parent bridge is what connects the students with their parents. The teacher and class entities have a one-to-many relationship because each class will have one teacher and teachers can have many classes. The class and student entities also have a one-to-many relationship because student is in one class and one class has many students. The student and parent entities have a many-to-many relationship because a student can have more than one parent, and parents can have more than one student. In order to demonstrate the many-to-many relationship between the students and parents the STUDENT_PARENT bridge is made that will have foreign keys from the parent and student entities.

Entities Analysis

CLASS_RW is the entity that stores information about the classes in Happy Learning. The information that is stored in this table includes class code, classroom number, phone, maximum capacity, directions, and the teacher code. Each class has one teacher. We have this in order to keep track of each class and always know where they are in the building and which staff is in charge of them.

Attribute Name	Attribute domain and size	Simple/Composite	Single/multi-value	Require/Optional	Unique/Du
CLASS_CODE_RW	CHAR(2)	Simple	Single	Required	Unique
ROOM_NUMBER_RW	VARCHAR(10)	Simple	Single	Required	Duplicated
PHONE_RW	CHAR(14)	Simple	Single	Required	Duplicated
MAX_CAPACITY_RW	TINYINT	Simple	Single	Required	Duplicated
DIRECTIONS_RW	VARCHAR(130)	Simple	Single	Optional	Duplicated
TEACHER_ID_RW	INT	Simple	Single	Required	Duplicated

The primary key is the class code and it is a simple key. The foreign key is TEACHER_ID_RW and it references the TEACHER_ID primary code in the TEACHER table.

The STUDENT_RW entity stores all the information on the students that are enrolled in the school. Information includes first and last name, and middle name and nickname if the have one. Their date of birth and address information will also be included. This way we know every single child that is enrolled and their personal information in order to contact parents if need be.

Attribute Name	Attribute domain and size	Simple/Composite	Single/multi-value	Required/Optional	Unique/Duplicated
STUDENT_ID_RW	INT	Simple	Single	Required	Unique
FIRST_NAME_RW	VARCHAR (22)	Simple	Single	Required	Duplicated
MIDDLE_NAME_RW	VARCHAR (22)	Simple	Single	Optional	Duplicated
LAST_NAME_RW	VARCHAR (22)	Simple	Single	Required	Duplicated
NICKNAME_RW	VARCHAR (20)	Simple	Single	Optional	Duplicated
DATE_OF_BIRTH_RW	DATE	Simple	Single	Required	Duplicated
STREET_RW	VARCHAR (35)	Simple	Single	Required	Duplicated
CITY_RW	VARCHAR (15)	Simple	Single	Required	Duplicated
ZIP_CODE_RW	CHAR(5)	Simple	Single	Required	Duplicated
CLASS_CODE_RW	CHAR(2)	Simple	Single	Required	Duplicated

The primary key is the STUDENT_ID_RW and it is a surrogate key because it is artificial. The foreign key is CLASS_CODE_RW and it refers to the primary key CLASS_CODE_RW in the CLASS_RW table.

The PARENT_RW entity stores important information on the parents who have children enrolled at Happy Learning. Information includes their names, different phone numbers, and emails. This way we can contact any parent in case of emergencies or updates and know who their child is.

Attribute Name	Attribute domain and size	Simple/Composite	Single/Multi-value	Required/Optional	Unique/Duplicated
PARENT_ID_RW	INT	Simple	Single	Required	Unique
FIRST_NAME_RW	VARCHAR (22)	Simple	Single	Required	Duplicated
MIDDLE_NAME_RW	VARCHAR (22)	Simple	Single	Optional	Duplicated
LAST_NAME_RW	VARCHAR (22)	Simple	Single	Required	Duplicated
HOME_PHONE_RW	CHAR(14)	Simple	Single		Duplicated

MOBILE_PHONE_RW	CHAR(14)	Simple	Single	Optional	Duplicated
WORK_PHONE_RW	CHAR(14)	Simple	Single	Optional	Duplicated
PERSONAL_EMAIL_RW	VARCHAR (30)	Simple	Single	Optional	Duplicated

The primary key is PARENT_ID_RW and it is a surrogate key because it is artificial and assigned to all parents in the system. There are no foreign keys because of the STUDENT_PARENT_RW bridge that is used to connect the other entity. Even though the phone attributes are optional, at least one of them must be filled in in order to properly contact parents.

The TEACHER_RW entity stores all the information on each teacher that we have at Happy Learning. The information included will be things such as name, phone number, email. And some information on their prior education in order to establish their credibility. We have this entity included to keep track of each of our staff members and ensure we have the credentials on account in order reference at any given moment.

Attribute name	Attribute domain and size	Simple/Composite	Single/multi-value	Required/Optional	Unique/Duplicated
TEACHER_ID_RW	INT	Simple	Single	Required	Unique
FIRST_NAME_RW	VARCHAR (22)	Simple	Single	Required	Duplicated
MIDDLE_NAME_RW	VARCHAR (22)	Simple	Single	Optional	Duplicated
LAST_NAME_RW	VARCHAR (22)	Simple	Single	Required	Duplicated
HOME_PHONE_RW	CHAR(14)	Simple	Single	Required	Duplicated
WORK_EMAIL_RW	VARCHAR (30)	Simple	Single	Required	Duplicated
PERSONAL_EMAIL_RW	VARCHAR (30)	Simple	Single	Optional	Duplicated
GRADUATED_COLLEGE_RW	VARCHAR (100)	Simple	Single	Required	Duplicated
HIGHEST_DEGREE_RW	VARCHAR (5)	Simple	Single	Required	Duplicated
DEGREE_AREA_RW	VARCHAR (25)	Simple	Single	Required	Duplicated

The primary key is TEACHER_ID_RW and it's a surrogate primary key because it is artificial. There are no foreign keys because the relationship is shown by TEACHER_ID_RW in

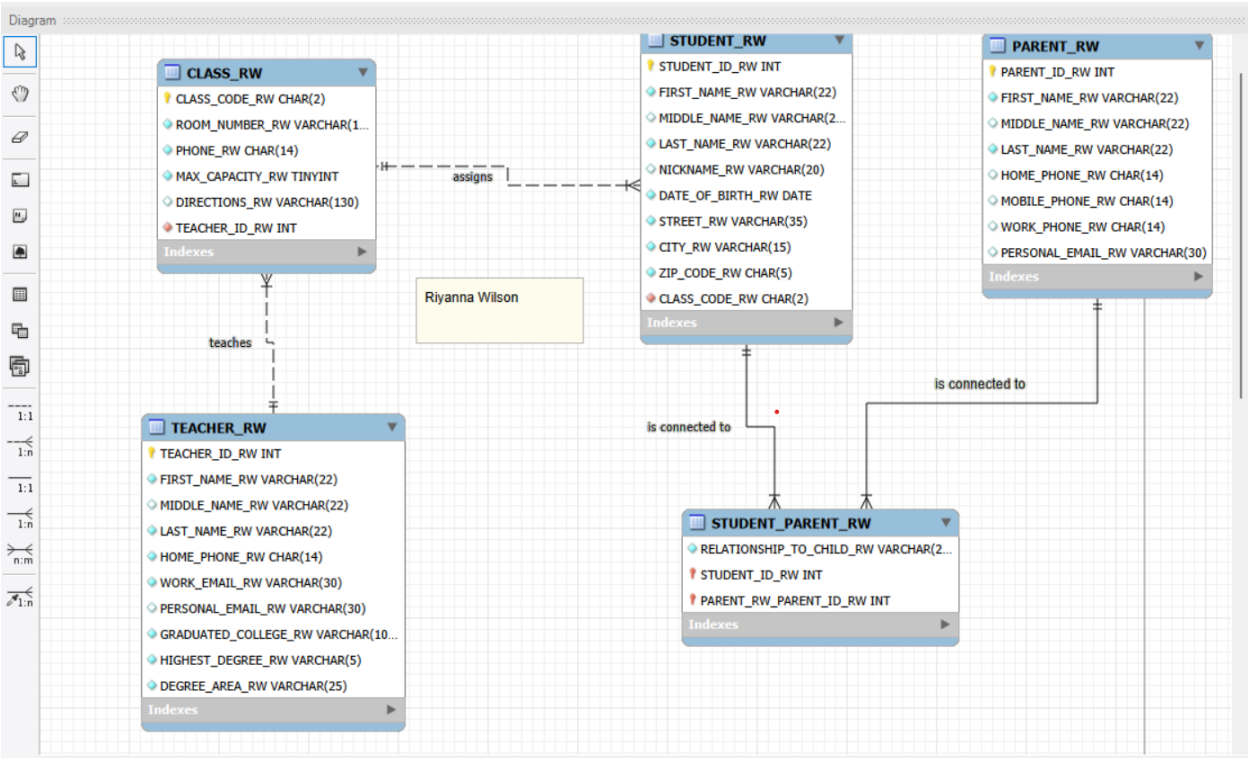
CLASS_RW. All teachers must have a bachelor's degree in order to teach at Happy Learning.

The STUDENT_PARENT_RW is a bridge entity that is used to connect the parents and students. We have this bridge because knowing which student belongs to which parent so we can contact them and let them know about their child's performance. The explanation of the relationship between the two will also be stored in this table

Attribute name	Attribute domain and size	Simple/composite	Single/multi-value	Required/optional	Unique/duplicated
STUDENT_ID_RW	INT	Simple	Single	Required	Duplicated
PARENT_ID_RW	INT	Simple	Single	Required	Duplicated
RELATIONSHIP_TO_CHILD_RW	VARCHAR(20)	Simple	Single	Required	Duplicated

The primary key is STUDENT_ID_RW and PARENT_ID_RW and it is a composite primary key because it is made up of two attributes. The foreign keys are STUDENT_ID_RW which references STUDENT_RW table and PARENT_ID_RW which references the PARENT_RW table.

Entity Relationship Diagram



Diagram

CLASS_RW - Table

Table Name: CLASS_RW Schema: **mydb**

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
CLASS_CODE_RW	CHAR(2)	☑	☑	☐	☐	☐	☐	☐	☐	
ROOM_NUMBER_RW	VARCHAR(10)	☐	☑	☐	☐	☐	☐	☐	☐	
PHONE_RW	CHAR(14)	☐	☑	☐	☐	☐	☐	☐	☐	
MAX_CAPACITY_RW	TINYINT	☐	☑	☐	☐	☐	☐	☐	☐	
DIRECTIONS_RW	VARCHAR(130)	☐	☐	☐	☐	☐	☐	☐	☐	
TEACHER_ID_RW	INT	☐	☑	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

Diagram

STUDENT_RW - Table

Table Name: STUDENT_RW Schema: **mydb**

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
STUDENT_ID_RW	INT	☑	☑	☐	☐	☐	☐	☐	☐	
FIRST_NAME_RW	VARCHAR(22)	☐	☑	☐	☐	☐	☐	☐	☐	
MIDDLE_NAME_RW	VARCHAR(22)	☐	☐	☐	☐	☐	☐	☐	☐	
LAST_NAME_RW	VARCHAR(22)	☐	☑	☐	☐	☐	☐	☐	☐	
NICKNAME_RW	VARCHAR(20)	☐	☐	☐	☐	☐	☐	☐	☐	
DATE_OF_BIRTH_RW	DATE	☐	☑	☐	☐	☐	☐	☐	☐	
STREET_RW	VARCHAR(35)	☐	☑	☐	☐	☐	☐	☐	☐	
CITY_RW	VARCHAR(15)	☐	☑	☐	☐	☐	☐	☐	☐	
ZIP_CODE_RW	CHAR(5)	☐	☑	☐	☐	☐	☐	☐	☐	
CLASS_CODE_RW	CHAR(2)	☐	☑	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

Diagram

TEACHER_RW - Table

Table Name: TEACHER_RW Schema: **mydb**

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
TEACHER_ID_RW	INT	☑	☑	☐	☐	☐	☐	☑	☐	
FIRST_NAME_RW	VARCHAR(22)	☐	☑	☐	☐	☐	☐	☐	☐	
MIDDLE_NAME_RW	VARCHAR(22)	☐	☐	☐	☐	☐	☐	☐	☐	
LAST_NAME_RW	VARCHAR(22)	☐	☑	☐	☐	☐	☐	☐	☐	
HOME_PHONE_RW	CHAR(14)	☐	☑	☐	☐	☐	☐	☐	☐	
WORK_EMAIL_RW	VARCHAR(30)	☐	☑	☐	☐	☐	☐	☐	☐	
PERSONAL_EMAIL_RW	VARCHAR(30)	☐	☐	☐	☐	☐	☐	☐	☐	
GRADUATED_COLLEGE_RW	VARCHAR(100)	☐	☑	☐	☐	☐	☐	☐	☐	
HIGHEST_DEGREE_RW	VARCHAR(5)	☐	☑	☐	☐	☐	☐	☐	☐	
DEGREE_AREA_RW	VARCHAR(25)	☐	☑	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

Diagram

PARENT_RW - Table

Table Name: PARENT_RW Schema: **mydb**

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
PARENT_ID_RW	INT	☑	☑	☐	☐	☐	☐	☑	☐	
FIRST_NAME_RW	VARCHAR(22)	☐	☑	☐	☐	☐	☐	☐	☐	
MIDDLE_NAME_RW	VARCHAR(22)	☐	☐	☐	☐	☐	☐	☐	☐	
LAST_NAME_RW	VARCHAR(22)	☐	☑	☐	☐	☐	☐	☐	☐	
HOME_PHONE_RW	CHAR(14)	☐	☐	☐	☐	☐	☐	☐	☐	
MOBILE_PHONE_RW	CHAR(14)	☐	☐	☐	☐	☐	☐	☐	☐	
WORK_PHONE_RW	CHAR(14)	☐	☐	☐	☐	☐	☐	☐	☐	
PERSONAL_EMAIL_RW	VARCHAR(30)	☐	☐	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

Diagram

STUDENT_PARENT_RW - Table

Table Name: STUDENT_PARENT_RW Schema: **mydb**

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
RELATIONSHIP_TO_CHILD_...	VARCHAR(20)	☐	☑	☐	☐	☐	☐	☐	☐	
STUDENT_ID_RW	INT	☑	☑	☐	☐	☐	☐	☐	☐	
PARENT_RW_PARENT_ID_RW	INT	☑	☑	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	